

We have the following subsequent corollary.

Corollary 1. *Let C^σ be an unbalanced signed cycle, then $p^+(C^\sigma) \geq 2$.*

Yu et al. [18] calculated the positive inertia of signed path P_n^σ .

Lemma 6. *Let P_n^σ be a signed path of order n , then $p^+(P_n^\sigma) = \lfloor \frac{n}{2} \rfloor$.*

Yu et al. [4, Theorem 3.1] characterized all connected signed graph G^σ with $p^+(G^\sigma) = 1$.

Lemma 7. *Let G^σ be a connected signed graph. Then $p^+(G^\sigma) = 1$ if and only if G^σ is a balanced complete multipartite signed graph.*

The length of a signed path (or simply a path) P^σ refers to the total number of edges in P^σ . For any two vertices y and z , the distance between them, denoted by $d(y, z)$, is defined as the length of the shortest path between y and z .

Lemma 8. *Let G^σ be a connected signed graph with girth g_r and let C^σ be a shortest cycle in G^σ . If $y, y' \in V(C^\sigma)$ and there exists a path P^σ of length k from y to y' satisfying $(V(P^\sigma) \setminus \{y, y'\}) \cap V(C^\sigma) = \emptyset$, then*

$$\left\lceil \frac{g_r}{2} \right\rceil \leq k.$$

Proof. Remember that C^σ have two different paths from y to y' . The total length of both the paths equals the length of C^σ , i.e., g_r . Thus, the shorter length of these paths is at most $\lfloor \frac{g_r}{2} \rfloor$. The shorter of these two paths from y to y' , followed by the path P^σ , forms a cycle of length at most $\lfloor \frac{g_r}{2} \rfloor + k$. Therefore, we obtain:

$$\left\lfloor \frac{g_r}{2} \right\rfloor + k \geq g_r,$$

and consequently

$$\left\lceil \frac{g_r}{2} \right\rceil \leq k.$$

This concludes the proof. □

Let $G^\sigma[H^\sigma]$ be a subgraph of G^σ induced by H^σ , and let x be a vertex outside H^σ . We denote the distance between x and $G^\sigma[H^\sigma]$ as: $d(x, G^\sigma[H^\sigma]) = \min\{d(x, y) \mid y \in H^\sigma\}$. Next, we define $N_j(G^\sigma[H^\sigma])$ as:

$$N_j(G^\sigma[H^\sigma]) = \{x \in V(G^\sigma) \setminus H^\sigma \mid d(x, G^\sigma[H^\sigma]) = j, \quad j = 1, 2, \dots, n\}.$$

The number of vertices in $N_j(G^\sigma[H^\sigma])$ is denoted by $|N_j(G^\sigma[H^\sigma])|$.

Lemma 9. *Let G^σ be a connected signed graph with girth g_r , and let C^σ be a shortest cycle in G^σ . If $p^+(C^\sigma) = p^+(G^\sigma)$, then $N_j(C^\sigma) = \emptyset$ for $j \geq 2$.*

Proof. Indeed, we only need to establish that $N_2(C^\sigma) = \emptyset$. Suppose, for the sake of contradiction, that $N_2(C^\sigma) \neq \emptyset$. Let $y' \in N_2(C^\sigma)$ and $y' \sim y \in N_1(C^\sigma)$. Then, y' is a pendant vertex of $G^\sigma[V(C^\sigma) \cup \{y, y'\}]$. Consequently,

$$p^+(G^\sigma) \geq p^+(G^\sigma[V(C^\sigma) \cup \{y, y'\}]) = p^+(C^\sigma) + 1 > p^+(C^\sigma),$$

by Lemmas 1 and 2, which contradicts the assumption that $p^+(C^\sigma) = p^+(G^\sigma)$. Hence, $N_2(C^\sigma) = \emptyset$, and therefore, $N_j(C^\sigma) = \emptyset$ for $j \geq 3$. This completes the proof. □

Smith [6] showed the following characterization.

Lemma 10. *A graph has precisely one positive eigenvalue if and only if the set of its non-isolated vertices constitute a complete multipartite graph.*